

SpecFirst React A11y

Phase 6 Documentation: Run Baseline Test

How the generated test suite proves the current
component fails the accessibility contract before
Bob is asked to remediate it

Phase 6 runs the generated Playwright tests against
the unpatched component. The goal is not to pass.
The goal is to prove the generated contract checks
catch real failures, and to record that baseline as the
starting point for Bob.

Prepared for the IBM Bob Hackathon workflow

1. Phase 6 in one sentence

Phase 6 executes the tests generated in Phase 5 against the unmodified component, classifies the outcome as baseline-red or unexpected-green, and writes a stable result artifact that Phase 7 uses to instruct Bob.

Pipeline context:

Phase 5: Generate tests Phase 6: Run baseline test ← you are here
Phase 7: Bob patches component Phase 8: Verify and report

The central design rule is that Phase 6 must prove the tests fail before Bob is allowed to patch. This prevents weak tests from silently passing against broken components.

2. Purpose

- Confirm the generated tests are strong enough to detect the accessibility gaps they were designed to catch.
- Establish a baseline record of which locked-spec checks passed and which failed before remediation.
- Give Phase 7 a structured list of failing check IDs to include in the Bob remediation prompt.
- Protect against weak tests by blocking the pipeline if all tests pass unexpectedly before any patching.
- Verify the generated artifacts have not been tampered with between Phase 5 and Phase 6.

3. Inputs

Input artifact	Role in Phase 6
testGenerationResult.json	Primary input. Must have status generated and nextPhase.canContinue true. Provides test file path, harness file path, locked spec hash, test file hash, harness hash, and the Playwright command.
lockedSpec.json	Hash is re-verified against testGenerationResult.lockedSpecHash to confirm the contract has not changed.
Generated test file (tests/a11y/*.spec.ts)	Hash is re-verified before running. Phase 6 refuses to run tests if the hash does not match the Phase 5 record.
Generated harness file	Hash is re-verified before running. If the harness was edited, Phase 6 produces status invalidated rather than infra-failed.
playwright.config.ts	Governs browser, base URL, and the Vite web server launch. Phase 6 relies on the config to start the demo app before running tests.

4. Outputs

Output artifact	Purpose
baselineResult.json	Machine-readable record of the baseline run. Status, hash snapshots, test summary, passed and failed check IDs, and nextPhase handoff. Primary input to Phase 7.
baselineOutput.json	Raw Playwright JSON reporter output. Preserved for debugging and evidence.
baselineOutput.txt	Raw stderr output from the Playwright process. Contains Vite server startup messages, error traces, and browser launch output.

5. Scope boundaries

Phase 6 should do:

- Read and validate testGenerationResult.json.

- Re-verify SHA-256 hashes of lockedSpec.json, the test file, and the harness file before running.
- Spawn `npx playwright test <testFile> --reporter=json` and capture stdout and stderr.
- Parse the JSON reporter output and map results back to locked-spec check IDs using the bracketed check ID pattern in test titles.
- Classify the outcome as red-confirmed, green-unexpected, infra-failed, or invalidated.
- Write baselineResult.json, baselineOutput.json, and baselineOutput.txt.
- Set `nextPhase.canContinue` true only on red-confirmed.

Phase 6 should not do:

- Patch the React component.
- Call Bob or generate a Bob prompt.
- Modify lockedSpec.json, the test file, or the harness.
- Regenerate tests.
- Claim accessibility compliance.
- Treat test assertion failures as infrastructure failures or vice versa.

6. Status model

Status	Meaning	nextPhase
red-confirmed	Playwright ran, at least one locked-spec check failed, and all artifact hashes match Phase 5 records. This is the expected successful baseline outcome for an unpatched component.	canContinue true
green-unexpected	All generated tests passed before any Bob patching. The component may already satisfy the contract, or the tests may be too weak. Pipeline pauses by default.	canContinue false
infra-failed	Tests did not run correctly due to a Vite server failure, browser launch failure, TypeScript compile error, or missing route. Not a test assertion failure.	canContinue false
invalidated	One or more artifact hashes do not match the Phase 5 record. A generated file was edited between Phase 5 and Phase 6. Do not run tests on tampered artifacts.	canContinue false
skipped	Phase 5 testGenerationResult.json status was not generated or nextPhase.canContinue was false.	canContinue false
failed	Unexpected Phase 6 implementation error, missing input file, or malformed JSON.	canContinue false

7. Hash verification before running

Phase 6 must verify three artifacts before spawning Playwright:

Artifact	Expected source	On mismatch
lockedSpec.json	testGenerationResult.lockedSpecHash	status: invalidated, do not run
Generated test file	testGenerationResult.testFileHash	status: invalidated, do not run
Generated harness file	testGenerationResult.harnessFileHash	status: invalidated, do not run

This step ensures that any manual edit between Phase 5 and Phase 6 is detected and

8. Infra failure versus assertion failure

A non-zero Playwright exit code alone does not mean infra-failed. Playwright exits non-zero when tests fail by design. Phase 6 must distinguish:

Condition	Classification
JSON reporter output is valid and contains test results with at least one failure	red-confirmed
JSON reporter output is valid and all tests passed	green-unexpected
JSON reporter output is empty or unparseable, and raw output contains server or browser errors	infra-failed
Playwright process itself could not be spawned	infra-failed
Any artifact hash mismatch detected before running	invalidated

9. Check ID traceability

The generated test titles follow the pattern:

```
accessible-name [accessible-name]  
keyboard-arrow-down-opens [keyboard-arrow-down-opens]
```

Phase 6 extracts the check ID from the bracketed suffix using the pattern `/([a-z0-9- ]+)$/`. This maps each Playwright result directly back to the locked-spec check it was generated from, ensuring the baseline failure list refers to contract check IDs, not raw test titles.

10. baselineResult.json structure

```
{
  "schemaVersion": "1.0.0",
  "phase": "baseline",
  "status": "red-confirmed",
  "runId": "2026-05-16T20-01-11-151Z-analyze",
  "lockedSpecPath": "specfirst/runs/<run-id>/lockedSpec.json",
  "lockedSpecHash": "sha256:...",
  "testFileHashBeforeRun": "sha256:...",
  "harnessFileHashBeforeRun": "sha256:...",
  "testSummary": { "total": 11, "passed": 6, "failed": 5, "skipped": 0 },
  "passedChecks": ["accessible-name", "combobox-role"],
  "failedChecks": [
    { "id": "keyboard-arrow-down-opens", "title": "...", "failureType":
      "assertion-failed", "message": "..." }
  ],
  "baselineValidation": { "testsRan": true, "testFileUnchanged": true,
    "harnessFileUnchanged": true,
    "lockedSpecUnchanged": true, "meaningfulFailureCount": 5, "redConfirmed":
      true },
  "nextPhase": { "canContinue": true, "bobPromptInput": "specfirst/runs/<run-
    id>/baselineResult.json" }
}
```

11. Failure cases

- testGenerationResult.json missing or status not generated.
- testGenerationResult.nextPhase.canContinue is false.
- lockedSpec.json, test file, or harness file does not exist.
- Any artifact hash mismatch produces invalidated immediately.
- Playwright process fails to spawn (path, permissions, browser not installed).
- Vite server fails to start within the configured timeout.
- JSON reporter output is empty or contains no test suites.

12. Handoff to Phase 7

Phase 7 (generate Bob prompt) can rely on baselineResult.json providing:

- status is red-confirmed.
- failedChecks contains structured entries with check IDs and failure messages.
- passedChecks lists checks that already pass and should not be disrupted by Bob's patch.
- lockedSpecHash, testFileHashBeforeRun, and harnessFileHashBeforeRun are

- `nextPhase.bobPromptInput` gives the path Phase 7 CLI accepts as input.

13. Definition of done

- Phase 6 reads `testGenerationResult.json` and refuses to proceed if status is not generated.
- All three artifact hashes are verified before Playwright runs.
- Playwright runs the generated test file and captures both JSON and raw output.
- Infra failures are distinguished from assertion failures.
- Failed check IDs are extracted and linked to locked-spec check IDs.
- `baselineResult.json`, `baselineOutput.json`, and `baselineOutput.txt` are written to the run directory.
- `nextPhase.canContinue` is true only on red-confirmed.
- Phase 6 does not patch code, call Bob, modify tests, or make compliance claims.

End of Phase 6 documentation.